

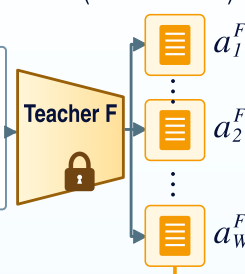
Cached teacher replies

1 Warm-Start

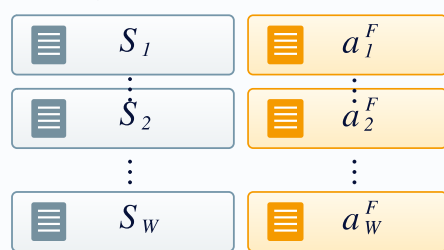
Early dialogue turns
(1 ... W)



Teacher replies
(cache and fix)



Warm-Start Pairs

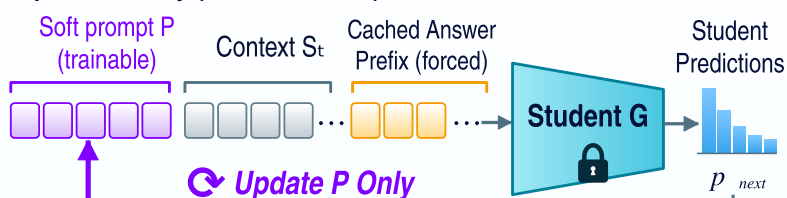


Cache Teacher Replies

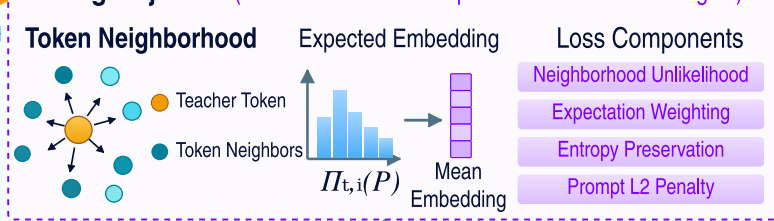
2 Soft-Prompt Mining

Discard
After Mining

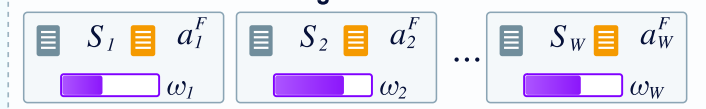
Input assembly (teacher-forced)



Mining Objective (teacher-forced student predictions vs. cached targets)



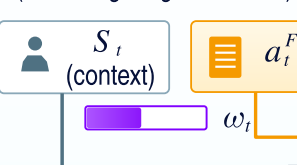
All Warm-Start Turns · Weights $\omega_t \propto r_t$



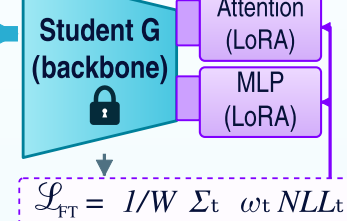
Optimize P · Freeze G · Anchoring Scores r_t

3 Local LoRA Distillation

All W warm-start turns
(anchoring weights $\omega_t \propto r_t$)



Student G (backbone)



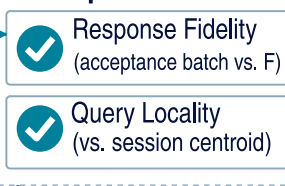
Adapted Student G_θ



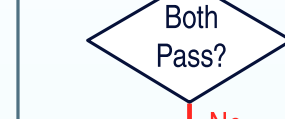
One-Time Adaptation

4 Gated Serving

Acceptance Checks



Both Pass?



Recheck

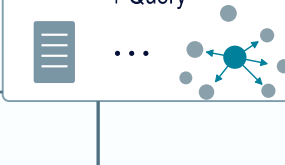


Refresh Summary + Centroid

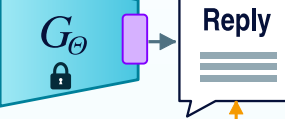


Prompt-Free Serving · No Online LoRA Updates

Summary + Recent K Turns
+ Query



Reply



Updated context



Recent queries

Legend: Frozen (lock icon), Offline update (purple box), Rollback (red dashed arrow)

